Chapter 8

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Chapter 8: Robotics for Beginners - "Robo Rebels Workshop" Worksheet Instructions: Yo, Robo Rebels! Welcome to the Chapter 8 worksheet, where you'll dive into the world of robotics and become creators of epic machines. This isn't just homework—it's your mission to brainstorm, design, and even plan a robot build that could race, prank, or solve real problems. It's short, action-packed, and designed for high schoolers like you, taking about 30-45 minutes. Grab a pen, some scrap paper, and your wildest ideas—let's make some bots that rule! ⚙️💥

Part 1: Quick Think - Robot Dream Machine (5 minutes) Task: Imagine a robot that could make your teen life easier (like doing homework or sneaking snacks). In 3-4 sentences, describe what it does and how you'd use ALL four STEM fields (Science, Technology, Engineering, Math) to make it real. Example: For a "SnackSneak Bot," Science studies stealth motion, Tech programs a quiet path, Engineering builds a small rolling frame, and Math calculates battery life for a 5-minute mission. **Your Robot Idea**: ___________________________ **Your STEM Breakdown**:

Part 2: Mini-Design Challenge - Pick Your Bot Battle (15 minutes) Task: Choose ONE robotics project from Chapter 8 below and sketch a quick design or idea. No need for perfection—just doodle on this sheet or extra paper and jot 2-3 notes on how STEM powers it up. Add a teen twist (like a cool name or social media feature).

- **Option A: Line-Follower Racer** - Design a bot to follow a black tape line for a class race. How does STEM make it stay on track?
- **Option B: Obstacle-Dodger Prankster** - Create a bot that avoids objects (like desks) to sneak around. How does STEM help it dodge?
- **Option C: Custom Helper Bot** - Build a bot for a small task (like carrying a phone charger). How does STEM solve your problem? **My Choice**: ___________________________ **Sketch Space**: (Draw here or attach a separate sheet) **STEM Notes**:
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Part 3: Code Crusher - Logic Blast (10 minutes) Task: Write a super simple "code plan" for your Part 2 bot. No real coding needed—just list 3-4 basic "if-then" steps it follows to do its job. Then, add ONE Math calc (like timing or distance). Example for Line-Follower: "If left sensor off line, turn left; if right sensor off, turn right; repeat until end; Math—10 cm/sec for 1m line = 10 sec." **My Bot's Code Plan**:

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My Math Calc: ____

Part 4: Robo Rebel Identity - Fun Wrap-Up (5 minutes) Task: Turn yourself into a Robotics Rebel based on this chapter! Pick a robot type (racer, prankster, helper) you vibed with most. Give yourself a rebel name and tagline tied to your robotics mission. Example: “Gear Grindr - Cranking Bots to Crush It!” **My Rebel Name**: ____________________________ **My Tagline**: ____________________________

Bonus Fun (Optional - 5 minutes) Robotics Meme Magic: Create a quick meme caption or hashtag about building or coding robots (like “#BotBossLife” or “Coding Glitches = Epic Fails!”). Write it below or share it in class for laughs and inspo! **My Meme Idea**: ____________________________

Reflection for the Win: What’s ONE idea or topic from Chapter 8 (like sensors or coding) that got you pumped to build a bot? Why does it matter to you or your squad?
